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Proposal Review 1 : 2601631

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Agency Name:	National Science Foundation
Agency Tracking Number:	2601631
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Multigraded Homological Algebra and Geometry
Rating:	Very Good

Review

Summary

Proposal Synopsis

Intellectual Merit Strengths

On a recent NSF Postdoc research fellowship, studied asymptotic properties of syzygies on products of projective spaces. Studied homology and regularity of graded modules of Cox ring: Toric varieties. Significant research was done on this fellowship. The broader impacts were also strong. The proposer organized conferences and organized math circles.

Proposes to study multigraded syzygies on stacks, understand section rings of line bundles on stacky curves and their syzygies, and their Np properties, especially for the canonical divisors. The proposer seeks to extend classical theorems for projective curves to stacky curves.

Study asymptotic syzygies of weighted projective spaces. Also proposes to develop software packages for computations with toric stacks in Macaulay2. Also seeks to understand asymptotic behavior of regularity of powers of graded ideals and ideals sheaves on a smooth projective toric variety with Cox ring S, especially in comparison with multigraded Seshadri constants.

Weaknesses

None

Rating for IM: VG

Broader Impacts

Strengths

Bruce proposes to organize conferences. They are mentoring two graduate students and several undergraduates.

Weaknesses

None

Rating for BI: VG

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

Summary Statement

The problems are interesting and it is expected that the investigator will make progress on them.

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